

EXECUTIVE MEMORANDUM

TO AMG Executive Team & Board of Directors
FROM Digital Transformation Advisory
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RE ERP Replacement Program - Strategy, Scope, Methodology & Data Continuity
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Executive Summary

Alpha Manufacturing Group faces an urgent and compounding operational and financial crisis. With approximately 80% of custom orders experiencing margin erosion, a fragile BOM synchronisation failure causing manufacturing delays, and no reliable contribution margin visibility per job, the legacy ERP is no longer a system of record; it is a liability that is actively destroying shareholder value.

Evidence from comparable programmes worldwide confirms that poorly executed ERP transformations amplify these problems rather than solve them. The Canada Phoenix Pay system, the UK NHS Lorenzo EHR programme, and the U.S. Air Force ERP consolidation all failed for the same reasons: weak governance, scope creep, inadequate data preparation, and insufficient change management. This programme is explicitly designed to avoid every one of those failure patterns.

This memorandum sets out the integrated strategic recommendation across eight areas:

- Why AMG must act now
- Lessons from real-world ERP failures and what they teach us
- Where AMG is today versus where it needs to be
- Overall program approach - principles, phasing, and risk management
- Scope and sequencing - Day-1 must-haves versus deliberate deferrals
- Implementation methodology - Vision through to Evolve
- Data migration and reporting continuity
- Recommended immediate next steps

The program is anchored to three non-negotiable principles: Operational Continuity First, Margin Recovery as the North Star, and Scope Discipline Above All Else. Every recommendation in this document traces back to one of these three principles.

Why AMG Must Act Now

AMG's current system landscape is not merely inefficient; it is causing measurable, ongoing financial harm. Three specific problems are driving the Board's mandate:

Problem	Business Impact
Margin Erosion	80% of custom orders experience cost overruns that were not anticipated at the time of pricing. With no real-time job costing, losses are discovered weeks after the job is shipped, too late to act.
BOM Synchronisation Failures	The Engineering BOM database syncs to the legacy ERP only once per night. When this sync fails, manufacturing receives outdated specifications, orders the wrong materials, incurs rework, and misses customer ship dates.
Blind Financial Reporting	Finance operates on Excel extracts from a 20-year-old system. Labour tracking is unreliable. There is no contribution margin view per job. Month-end close is slow, and the numbers produced are not fully trusted by leadership.

The compounding nature of these problems is critical to understand. BOM errors inflate material costs. Unreliable labour tracking means cost overruns are invisible. And because there is no job-level profitability view, the estimating process never gets corrected, and the same errors feed into the next job. This is why 80% of orders are affected, not an isolated few.

Lessons From Real-World ERP Failures

Before prescribing a solution, it is essential to understand what causes ERP programmes of this scale to fail. The evidence is unambiguous:

Programme	Cost Reality	Lesson for AMG
UK NHS - Lorenzo EHR	Planned £6.4B → Actual £10B. Cancelled 2011.	Scale amplifies mistakes; integration failures can sink the entire programme.
Canada Phoenix Pay - Oracle	\$3.5B CAD and rising. 80% payroll errors.	Payroll modernisation requires meticulous testing; errors destroy trust instantly.
Queensland Health - SAP/IBM	Planned \$6M AUD → Actual \$1.2B AUD.	Excessive customisation and weak governance are a guaranteed recipe for failure.
U.S. Air Force - Oracle/CSC	\$1B+ over 7 years. Never went live.	Unrealistic scope and no vendor accountability guarantee programme failure.
U.S. Virtual Fence - Boeing	\$1.4B for 23 miles. Cancelled.	Technology must match operational reality; physical and digital must align.

Six Common Threads Behind Every Failure

- **Politically Driven Procurement:** vendor selection decisions made for the wrong reasons introduce risk from Day 1.
- **Inadequate Requirements Gathering:** without thorough requirements, project expectations are never clearly defined.

- **Weak Governance:** without strong oversight and clear decision authority, scope and costs spiral.
- **Inadequate Change Management:** technology adoption is a human challenge first; people resist change without support.
- **Unrealistic Budgets and Timelines:** overpromising at the start leads to underdelivering later and shortcuts that cause failures.
- **No Vendor Accountability:** open-checkbook supplier relationships rarely end well; contracts must define deliverables.

Where AMG Is vs. Where It Needs to Be

Where AMG Is Today	Where AMG Needs to Be
Manual, complex processes riddled with inefficiencies specific to each location	Standardised, end-to-end processes enabling efficiency, consistency, and scale
Disparate applications hindering reporting, insights, and analytics	Unified platform with real-time reporting and contribution margin visibility per job
Limited data model and poor master data governance across 5 locations	Strong data governance anchored to a single, trusted source of truth
Task-focused workforce with limited digital capabilities	Digitally capable workforce driving innovation and freeing capacity for value-add activities
Fragile BOM sync causing manufacturing delays and compounding margin erosion	Integrated Engineering and ERP on one system, eliminating BOM errors from Day 1

The transformation is not simply a technology replacement; it is a fundamental change to how AMG operates. The ERP is the enabler; the business change is the objective. This distinction matters enormously for change management and user adoption.

Overall Program Approach

Three Non-Negotiable Guiding Principles

✓ Operational Continuity First

No disruption to order fulfilment, customer commitments, or monthly financial close. Every scope and timing decision is evaluated against this principle first. Cutover only happens after parallel runs confirm full parity with the legacy system. The business cannot stop to implement a system.

✓ Margin Recovery as the North Star

The entire programme exists to stop 80% of orders losing money. Every module selected for Day-1 go-live must contribute directly to contribution margin visibility. Job costing is not optional; it is the primary deliverable of the transformation. Leadership must be able to see order profitability from the very first day the new system is live.

✓ Scope Discipline Above All Else

Fewer than 15 customisations at go-live. Every scope addition must pass a three-question test: Is it required for Day-1 operations? Does a standard configuration already exist? What is the ongoing maintenance cost? The Executive Steering Committee holds veto power over all scope changes. The CFO and COO must personally approve any addition.

Four-Phase Delivery Roadmap

Phase & Timing	Scope & Deliverables
Phase 0: Foundation (Months 1-2)	ERP vendor selection and contracting; mandatory BOM audit and data cleanse; project governance and Steering Committee established; parallel reporting layer stood up; change management plan developed.
Phase 1: Core Go-Live (Months 3-12)	Finance and multi-entity consolidation; unified Engineering and BOM management; Manufacturing execution and MRP; Project and job costing; Procurement and multi-location inventory. Single cutover event across all 5 locations.
Phase 2: Stabilise (Months 12-15)	Hypercare and issue resolution; KPI dashboard validation; user adoption measurement; process improvement cycle; CRM integration scoping.
Phase 3: Extend (Month 15+)	CRM integration; HRIS and Payroll migration; advanced analytics and BI suite; EDI and customer portal automation; continuous improvement programme.

Top 5 Program Risks and Mitigations

Risk	Severity	Mitigation
BOM Data Quality	HIGH	Mandatory BOM audit in Phase 0; Engineering sign-off required before migration; BOM freeze during cutover window.
Scope Creep	HIGH	Executive Steering Committee with veto power; formal change control process; CFO + COO approval for all scope additions.
Key Person Dependency	MEDIUM	Structured knowledge transfer in Phase 0; shadow training programme; strong vendor hypercare SLA negotiated at contract stage.
Cutover Integration Failure	HIGH	Parallel running for minimum 2 pay periods and 1 full month-end close before legacy decommission; pre-defined rollback trigger.
User Adoption Failure	MEDIUM	Change management programme starts Month 1; role-based training; super-user network established at all 5 locations.

Scope & Sequencing

Day-1 Must-Haves

The Five Modules Required at Go-Live

1. **Financial Accounting & Multi-Entity Consolidation:** General ledger, AP/AR, multi-currency across US and Canadian entities. The company cannot legally operate without this.
2. **Unified Engineering & BOM Management:** The single highest-value capability. Eliminates the fragile nightly sync that is the root cause of manufacturing delays and material cost overruns. This is the #1 priority.
3. **Manufacturing Execution & MRP:** Work orders, production scheduling, shop floor data capture, and labour actuals. Manufacturing is AMG's core business — it cannot operate without system support.
4. **Procurement & Multi-Location Inventory Management:** Purchase orders, vendor management, stock levels across all 5 locations. Manufacturing cannot work without materials; materials cannot arrive without procurement.
5. **Project & Job Costing:** The module that directly delivers the Board's mandate. Tracks actual costs per job against estimates in real time. If this is not live on Day 1, the entire programme has failed its primary objective.

Deliberate Day-1 Exclusions

The following are explicitly out of scope for Phase 1. Including any of them would jeopardise the 12-month target. The business can continue with current arrangements for each of these during the stabilisation period:

Excluded Item	Rationale & When It Is Addressed
CRM Integration	The cloud CRM is already working, and sales are being made. Integration adds cutover risk without Day-1 operational value. Addressed in Phase 3.
HRIS & Payroll Migration	Payroll is a zero-tolerance disruption area. A payroll failure has immediate, severe consequences. Migrate only after core ERP is stable. Addressed in Phase 3.
Advanced BI & Analytics Suite	Base KPI reporting (contribution margin, BOM accuracy, on-time delivery) goes live Day 1. Full analytics suite requires clean historical data to be meaningful. Phase 3 deliverable.
EDI & Customer Portal Automation	Current manual processes continue. Adding integration complexity at go-live introduces failure risk for no immediate operational gain. Addressed in Phase 3.
Full Historical Data Migration (7+ Years)	20 years of legacy data is messy and partially corrupted. Migrating all of it would extend the timeline significantly. 2–3 years of active data migrates; the remainder stays in a read-only legacy archive.

Customisation Discipline

- Target: fewer than 15 customisations at go-live. Every customisation delay implementation breaks during vendor upgrades and creates permanent maintenance cost.
- All customisation requests must pass a three-question test before approval:
 1. Is this genuinely required for Day-1 operations?
 2. Does a standard configuration option already exist?
 3. What is the annual maintenance cost?
- Acceptable customisations: engineer-to-order BOM complexity and multi-entity consolidation where no standard equivalent exists. Unacceptable: UI preferences, reports in existing formats, legacy system behaviour replication.
- The principle is: adopt the software, do not adapt the software. The ERP's standard processes are built on best practices from thousands of companies. Where AMG differs, the question must be asked: is our way genuinely better, or merely familiar?

Implementation Methodology

The programme follows a structured five-phase implementation methodology. Each phase has formal sign-off gates. No phase begins until the prior phase's deliverables have been reviewed, approved, and baselined.

Phase	Key Activities & Deliverables
Vision	Define transformation vision and objectives; project kick-off and mobilisation; conduct cross-departmental workshops to validate business requirements; develop change management plan; provision ERP system instances.
Validate	Conduct gap analysis between AMG's requirements and ERP standard processes; develop final solution architecture; solution review and sign-off by Steering Committee; conduct Conference Room Pilot 1 (CRP1); review and formally sign off all validated requirements.
Construct & Test	Configure applications for User Acceptance Testing (UAT); develop data migration plan and begin extraction and cleansing; prepare functional and technical specifications for any approved customisations; develop training manuals; conduct end-user training sessions; review and sign-off training.
Deploy	Execute full data migration to production; conduct UAT and formal sign-off; deploy cutover production plan; establish change management tools for go-live; execute go-live readiness checks; Go-Live.
Evolve	Post go-live hypercare support; establish continuous improvement programme; transition to managed service; validate KPIs and reporting accuracy; measure user adoption; begin scoping Phase 3 extensions.

Key Methodology Principles

- Conference Room Pilots (CRP1 and CRP2) are mandatory. These are structured sessions where AMG's business users test the configured system against real scenarios before UAT begins.
- UAT must be signed off by business users and not by the IT or project team. The people who will use the system daily must confirm it works for their role before go-live is approved.
- The go-live readiness checklist is a formal gate. Every item must be green before the cutover begins. There is no negotiating on go-live criteria under deadline pressure.

Data Migration & Reporting Continuity

Historical Data Migration Strategy

Not all data needs to move to the new system. AMG's data falls into distinct categories that are treated differently based on operational necessity and migration risk:

Data Category	Migration Approach
Open Sales Orders & Active BOMs	Full migration to new ERP. Manually validated by Engineering and Sales before cutover. These are live business commitments and cannot be missed.
Open Purchase Orders & Commitments	Full migration. Procurement team validates all open POs before the cutover window begins.
GL Balances at Period-End Cutover Date	Period-end balances migrated as a clean starting point. Full GL transaction history stays in legacy system (read-only).
2-3 Years of Transactional History	Migrated to new ERP or parallel data warehouse. Sufficient for operational queries and management reporting without the risk of migrating full legacy data.
7+ Year Statutory Archive	Retained in read-only legacy system or exported to an immutable archive. Legal obligation is accessibility, not migration. No migration required.
Employee & Payroll History	Remains in HRIS. Not touched in Phase 1. Migration assessed during Phase 3 scoping.

Reporting Continuity During Transition

- **Months 1-12 (Pre-Cutover):** Legacy ERP remains the system of record. A parallel reporting layer (e.g. Power BI connected to the legacy system) provides consistent KPI views as data is progressively cleansed. Leadership does not go blind during the project.
- **Cutover Period:** A two-week dual-entry window ensures no transactions are lost between systems. All transactions are entered into both systems simultaneously during this window.
- **Post-Cutover:** New ERP becomes the system of record. Legacy is read-only for a minimum of 6 months before archiving and decommission.
- **Day-1 Executive KPI Dashboard:** Contribution margin by job, on-time delivery rate, and BOM accuracy rate go live on Day 1 of the new ERP. This is a programme success criterion, not a Phase 3 deliverable.

Month-End Close Protection Plan

Four-Layer Protection for Financial Close Continuity

- **Cutover Timing:** Go-live is scheduled for the first business day of a calendar month, never mid-month. This ensures the entire first trading period sits cleanly in the new system, with no transactions split across two systems.

- **Parallel Close:** Finance runs a full month-end close in both systems simultaneously for the first post-go-live period. The two sets of accounts must reconcile to within \$1,000 variance before the new system is signed off as the official record.
- **Finance Hypercare:** ERP vendor finance specialists are embedded on-site for the first two close cycles. The vendor SLA guarantees a 2-hour response during close week. No remote support; on-site presence is non-negotiable.
- **Rollback Trigger:** A pre-defined rollback criterion is agreed before go-live. If month-end close cannot be completed within 5 business days post-cutover, the programme automatically reverts to the legacy system until the root cause is identified and resolved.

Conclusion & Recommended Next Steps

AMG has a narrow and time-sensitive window to execute this transformation before the operational and financial risks of the legacy system become unmanageable. Every month of delay is another month of margin erosion on 80% of custom orders, another month of manufacturing disruptions from BOM failures, and another month of leadership making decisions without reliable financial data.

The programme structure in this memorandum has been deliberately designed to deliver a successful go-live within the Board's 12-month mandate while protecting the business from the failure patterns that have derailed comparable programmes worldwide. Every decision on phasing, scope, methodology, and data traces back to the three non-negotiable principles: Operational Continuity First, Margin Recovery as the North Star, and Scope Discipline Above All Else.

Immediate Actions Required

Timing	Action Required
Week 1	Board formally authorises the programme. C-suite Program Sponsor appointed. Project Manager engaged. Executive Steering Committee constituted with veto authority over scope.
Weeks 2-4	RFP issued to 3–5 cloud ERP vendors. RFP must include AMG's engineer-to-order BOM requirements, multi-entity consolidation needs, and the 12-month go-live constraint as non-negotiable selection criteria.
Month 2	Vendor selected and contracted. Phase 0 BOM audit begins immediately. Data cleanse programme initiated. Change management planning starts.
Month 3	Full programme kick-off. All legacy system customisations frozen immediately; no new changes to the old system from this point forward.